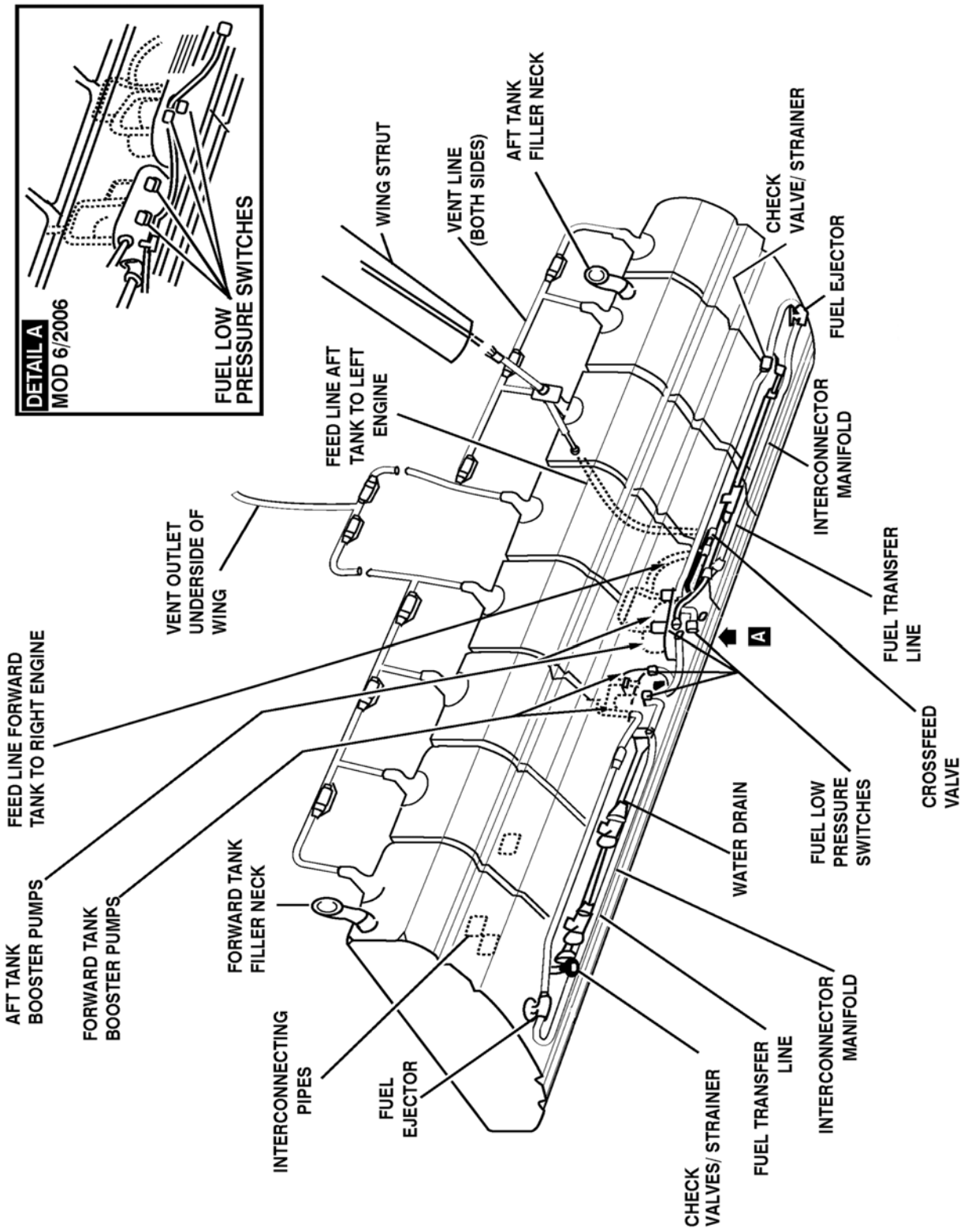
Fuel (With Mod S.O.O. 6247) – General Data

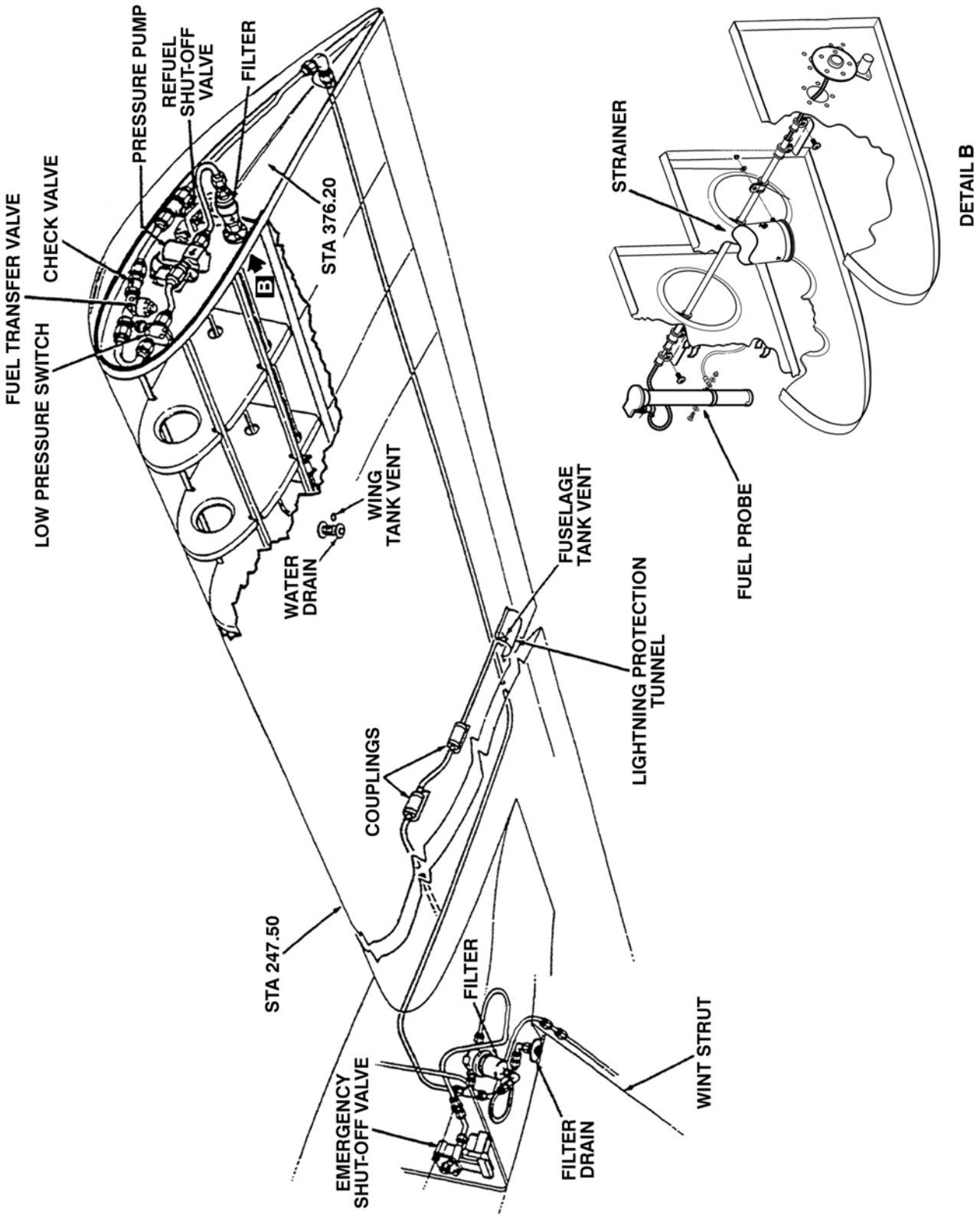
1. General

Fuel is contained in four fuel tanks, two in the fuselage and two (Mod S.O.O. 6247 and Mod 6/2046) in the wings. The forward fuselage tank and the right wing tank supply fuel to the right engine, the aft fuselage tank and left wing tank supply the left engine. A crossfeed valve when operated provides a fuel supply to both engines from one fuselage tank and associated wing tank. The fuselage tanks, each consisting of four interconnected fuel cells, deliver fuel under booster pump pressure to a tee union on each nacelle mounted fuel filter. The wing tanks, if installed, form an integral part of each wing, and deliver fuel under pump pressure to the same tee unions as the fuselage tanks. From the tee union on each fuel filter, fuel is routed through a fuel emergency shut-off valve and to the engine. A fuel flow transducer is installed in the fuel line between the FCU and the fuel flow divider. To permit checks for water accumulation in the fuel system, water drain valves are provided in each wing tank, in the forward and aft fuselage tank interconnector manifolds, and in each fuel filter. In addition, a fuselage tank fuel drain valve is provided in each sump plate for No. 4 and No. 5 collector cells. A fuel quantity indicating system and a fuel low pressure indicating system is provided for each tank. A fuel low level indicating system is provided for each fuselage fuel tank. Refer to Figure 1 and Figure 2 for details.



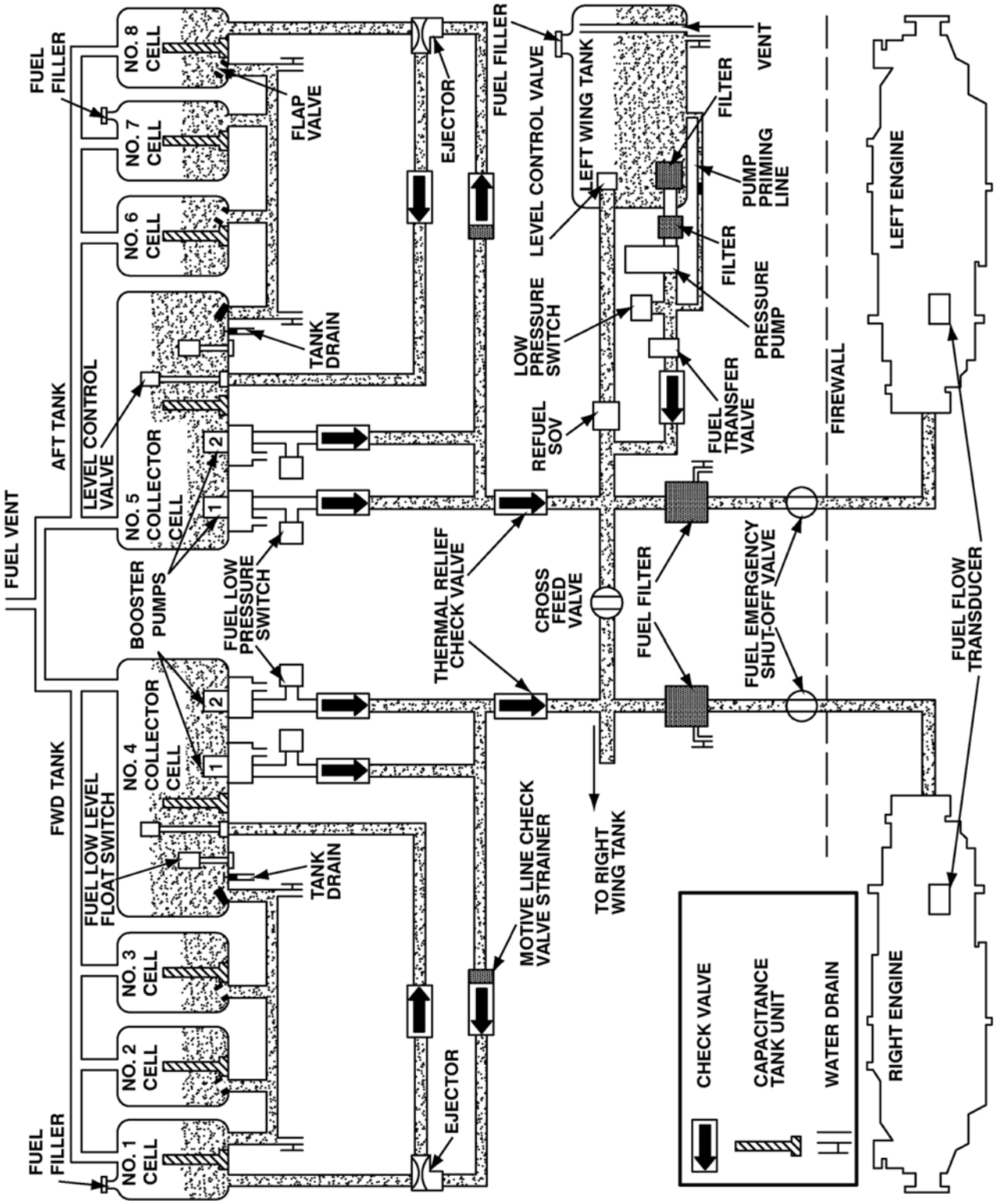
Fuel System Arrangement
Figure 1 (Sheet 1 of 2)

Print Date: 2025-09-15



Fuel System Arrangement
Figure 1 (Sheet 2 of 2)

Print Date: 2025-09-15



Fuel System Schematic
Figure 2

Print Date: 2025-09-15

2. Adjustment/Test

Note Aircraft must be reasonably level, and in the open air, during all fuel system checks. Wing tanks and components are checked separately to permit testing of fuselage tank system if wing tanks are not to be used.

A. Electrical Check (No Fuel)

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Ensure that fuel tanks are drained of all usable fuel and, with all relevant circuit breakers engaged, set switches in modes 1 to 5 given in Table 1 — Fuel System Electrical Check – No Fuel. Check that all conditions are produced as shown.

B. Transfer and Amber Block Message Check

1) On the fuel control panel, verify FWD BOOST PUMP 1 switch is in the OFF position and FWD BOOST PUMP 2 switch in the NORMAL position.

2) On the fuel control panel, verify AFT BOOST PUMP 1 switch is in the OFF position and AFT BOOST PUMP 2 switch is in the NORMAL position.

3) Place 30 Imperial (36 U.S.) gallons of fuel in forward fuselage tank and allow fuel level to stabilize for 5 minutes.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), engage relevant circuit breakers and observe that the forward boost pump amber blocks Pmp1 and Pmp2 are displayed and the Fwd Fuel Qty Low caution message is displayed.

Note Fuel in collector cell is below operating level of low level float switch.

5) On fuel control panel, set FUEL SELECTOR rotary switch to NORMAL and select forward fuselage tank BOOST PUMP 1 switch to the ON position.

6) Check that both forward boost pump amber block Pmp1 and Pmp2 messages are no longer displayed and Fwd Fuel Qty Low caution message is no longer displayed.

Note Fuel transfer takes place, raising fuel level in collector cell to operate low level float switch.

7) Switch off booster pump and verify Fwd Fuel Qty Low caution message is no longer displayed.

8) Repeat Para 2.B.3) through Para 2.B.7), substituting aft fuselage tank for forward.

C. Pressure Check

1) Ensure that each fuselage tank contains a minimum of 30 Imperial (36 U.S.) gallons of fuel.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.) and engage relevant circuit breakers.

3) Position fuel levers in overhead console to OFF and FUEL SELECTOR rotary switch on fuel control panel to NORMAL. Verify that the fuel pump amber blocks Pmp1 and Pmp2 for the forward and aft fuel tank are displayed.

4) On fuel control panel, select FWD and AFT BOOST PUMP 1 switches to ON. Verify that the fuel pump amber blocks Pmp1 and Pmp2 for the forward and aft tank on the fuel pane are no longer displayed.

5) Inspect complete fuselage fuel system for leaks. Repair leaks before proceeding with pressure check.

6) Select and hold booster pump switches to TEST. Check that both No.1 boost pump Pmp1 amber block messages are displayed and both No. 2 boost pump Pmp2 amber block messages are not displayed.

7) Release booster pump switches to OFF and check that all four fuel pump Pmp1 and Pmp2 amber block messages are displayed.

D. Function Check

1) Disconnect each engine fuel supply flexible hose, routed between emergency fuel shut-off valve on firewall and engine oil-to-fuel heater at elbow on heater. Place disconnected end of hose in a suitable container to receive fuel.

2) Check that forward fuselage tank contains approximately 30 Imperial (36 U.S.) gallons of fuel.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), engage relevant circuit breakers and, using Table 2 — Fuel System Function Check – Fuel In Forward Tank, check fuel system in modes 1 to 5.

4) With switches in mode 2 and fuel flow through engine supply flexible hose stabilized, measure and record time taken for one Imperial (1.2 U.S.) gallon of fuel to be pumped from forward fuselage tank. Maximum time allowed is 30 seconds.

5) With switches in mode 5 and fuel flow through engine supply flexible hose stabilized, measure and record time taken for one Imperial (1.2 U.S.) gallon of fuel to be pumped from forward fuselage tank. Maximum time allowed is 30 seconds.

6) Drain forward fuselage tank and check that aft fuselage tank contains approximately 30 Imperial (36 U.S.) gallons of fuel.

7) Repeat Para 2.D.3) through Para 2.D.5) for aft fuselage tank, substituting AFT for FWD and Table 3 — Fuel System Function Check – Fuel in Aft Tank for Table 2 — Fuel System Function Check – Fuel In Forward Tank.

8) Ensure that boost pump switches are off and remove external power.

9) Reconnect engine fuel supply flexible hoses to engine oil-to-fuel heaters.

E. Operational Test (Wing Tanks)

CAUTION AVOID DRY RUNNING OF WING TANK PUMPS.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.) and engage relevant circuit breakers.

2) Fill forward and aft fuselage tanks and ensure that wing tanks are empty.

3) Select the ENGINE switch position on the left wing fuel tank switch on the fuel tank control panel and check that the associated caution message Left Wing Fuel Pump is displayed.

4) Verify the left fuel pump amber block Pump message is displayed.

5) Select OFF at left wing fuel tank switch.

6) Repeat Para 2.E.3) and Para 2.E.5) for right wing fuel tank.

7) Select REFUEL on both wing fuel tank switches and NORMAL at FUEL SELECTOR rotary switch on fuel control panel.

8) Verify that the wing tank advisory CAS messages Left Wing Refuel On and Right Wing Refuel On are displayed.

9) Position fuel levers in overhead console to OFF and select fuselage tank BOOST PUMP 1 switches to ON.

10) Observe wing tank fuel quantity indications and check that tanks start to fill and then stop automatically at FULL. Check that no further spillage occurs at vent holes in lower wing skin.

11) Select OFF on wing fuel tank switches and switch off fuselage tank booster pumps.

12) Select the ENGINE switch position on the left and right wing fuel tank control panel and check that the associated caution messages Left Wing Fuel Pmp and Right Wing Fuel Pmp are not displayed. Verify the left and right wing tank fuel pump blocks Pump are now displayed green. Return both switches to OFF and remove external power supply.

F. Fuel Flow Test (Wing Tanks)

1) Disconnect left wing fuel tank pressure supply line from tee at nacelle-mounted fuel filter. Cap open port in tee.

2) Connect a length of hose to disconnected end of pressure supply line and place end of hose in a suitable container to receive fuel.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), engage relevant circuit breaker, and select ENGINE at appropriate switch on wing fuel tanks control panel.

4) With fuel flow through hose stabilized, measure and record time taken for 1 Imperial (1.2 U.S.) gallon of fuel to be pumped from left wing tank. Maximum time allowed is 85 seconds.

5) Select left wing fuel tank switch OFF.

6) Remove hose from left wing fuel tank pressure supply line and blanking cap from tee on fuel filter. Reconnect fuel line to tee.

7) Repeat Para 2.F.1) through Para 2.F.6) for right wing fuel tank.

8) Ensure all switches are off and remove external power supply.

CAUTION **DO NOT ALLOW PUMPS TO RUN MORE THAN 30 SECONDS BEFORE RETURNING THE SWITCHES TO THE OFF POSITION.**

Table 1 Fuel System Electrical Check – No Fuel

SWITCHES	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	NORMAL	BOTH ON FWD	BOTH ON AFT	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	NORMAL	FUEL OFF	FUEL OFF	FUEL OFF	FUEL OFF
Fuel Emergency Shut-off – Right	NORMAL	FUEL OFF	FUEL OFF	FUEL OFF	FUEL OFF
Fwd Boost Pump 1	OFF	ON	ON	TEST	OFF
Aft Boost Pump 1	OFF	ON	ON	TEST	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	EMERGENCY

Table 1 Fuel System Electrical Check – No Fuel (continued)

SWITCHES	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
PRODUCED CONDITION	↓	↓	↓	↓	↓
No. 1 Boost Pump Fwd	OFF	ON	OFF	OFF	OFF
No. 2 Boost Pump Fwd	OFF	ON	OFF	ON	ON
No. 1 Boost Pump Aft	OFF	OFF	ON	OFF	OFF
No. 2 Boost Pump Aft	OFF	OFF	ON	ON	ON
Crossfeed Valve	CLOSED	OPEN	OPEN	CLOSED	CLOSED
Shut-off Valve – Left	OPEN	CLOSED	CLOSED	CLOSED	CLOSED
Shut-off Valve – Right	OPEN	CLOSED	CLOSED	CLOSED	CLOSED
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	ON	ON	ON	ON	ON
Fwd Pmp2	ON	ON	ON	ON	ON
Aft Pmp1	ON	ON	ON	ON	ON
Aft Pmp2	ON	ON	ON	ON	ON
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	ON	ON	OFF	ON	ON
Boost Pump 2 Fwd Press	ON	ON	OFF	ON	ON
Boost Pump 1 Aft Press	ON	OFF	ON	ON	ON
Boost Pump 2 Aft Press	ON	OFF	ON	ON	ON
WARNING MESSAGE STATUS					
Fuel Low Both Tanks	ON	ON	ON	ON	ON

CAUTION DO NOT ALLOW PUMPS IN EMPTY TANKS TO RUN MORE THAN 30 SECONDS.

Table 2 Fuel System Function Check – Fuel In Forward Tank

SWITCHES	FORWARD TANK				
	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	BOTH ON FWD	NORMAL	BOTH ON FWD	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fuel Emergency Shut-off – Right	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fwd Boost Pump 1	OFF	ON	ON	TEST	OFF
Aft Boost Pump 1	ON	OFF	OFF	OFF	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	BOTH TO EMERGENCY
PRODUCED CONDITION	↓	↓	↓	↓	↓
Full Bore Flow At Left Engine	NO FLOW	NO FLOW	YES	NO FLOW	NO FLOW
Full Bore Flow At Right Engine	NO FLOW	YES	YES	NO FLOW	YES
Fuel Quantity Indication – Fwd	NO CHANGE	CHANGE	CHANGE	NO CHANGE	CHANGE
Fuel Quantity Indication – Aft	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	OFF	OFF	OFF	ON	ON
Fwd Pmp2	OFF	OFF	OFF	OFF	OFF
Aft Pmp1	ON	ON	ON	ON	ON
Aft Pmp2	ON	ON	ON	ON	ON
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	OFF	OFF	OFF	ON	ON
Boost Pump 2 Fwd Press	OFF	OFF	OFF	OFF	OFF
Boost Pump 1 Aft Press	OFF	ON	OFF	ON	ON
Boost Pump 2 Aft Press	OFF	ON	OFF	ON	ON
Fwd Fuel Qty Low	OFF	OFF	OFF	OFF	OFF
Aft Fuel Qty Low	ON	ON	ON	ON	ON

CAUTION **DO NOT ALLOW PUMPS IN EMPTY TANKS TO RUN MORE THAN 30 SECONDS.**

Table 3 Fuel System Function Check – Fuel in Aft Tank

SWITCHES	AFT TANK				
	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	BOTH ON AFT	NORMAL	BOTH ON AFT	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fuel Emergency Shut-off – Right	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fwd Boost Pump 1	ON	OFF	OFF	OFF	OFF
Aft Boost Pump 1	OFF	ON	ON	TEST	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	BOTH ON EMERGENCY
PRODUCED CONDITION	↓	↓	↓	↓	↓
Full Bore Flow At Left Engine	NO FLOW	YES	YES	NO FLOW	YES
Full Bore Flow At Right Engine	NO FLOW	NO FLOW	YES	NO FLOW	NO FLOW
Fuel Quantity Indication – Fwd	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE
Fuel Quantity Indication – Aft	NO CHANGE	CHANGE	CHANGE	NO CHANGE	CHANGE
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	ON	ON	ON	ON	ON
Fwd Pmp2	ON	ON	ON	ON	ON
Aft Pmp1	OFF	OFF	OFF	ON	ON
Aft Pmp2	OFF	OFF	OFF	OFF	OFF
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	OFF	ON	OFF	ON	ON
Boost Pump 2 Fwd Press	OFF	ON	OFF	ON	ON
Boost Pump 1 Aft Press	OFF	OFF	OFF	ON	ON
Boost Pump 2 Aft Press	OFF	OFF	OFF	OFF	OFF

Table 3 Fuel System Function Check – Fuel in Aft Tank (continued)

SWITCHES	AFT TANK				
	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fwd Fuel Qty Low	ON	ON	ON	ON	ON
Aft Fuel Qty Low	OFF	OFF	OFF	OFF	OFF

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